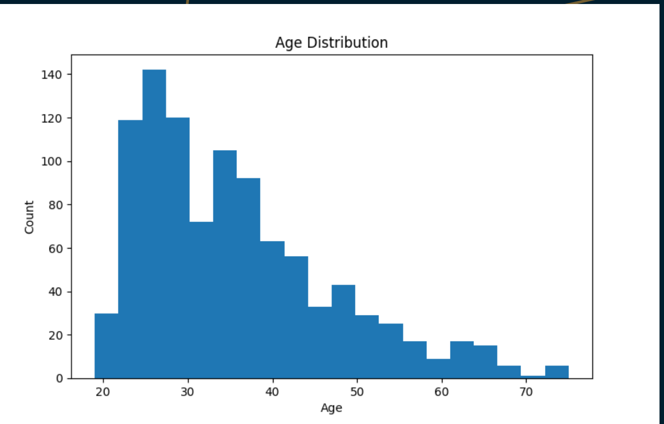
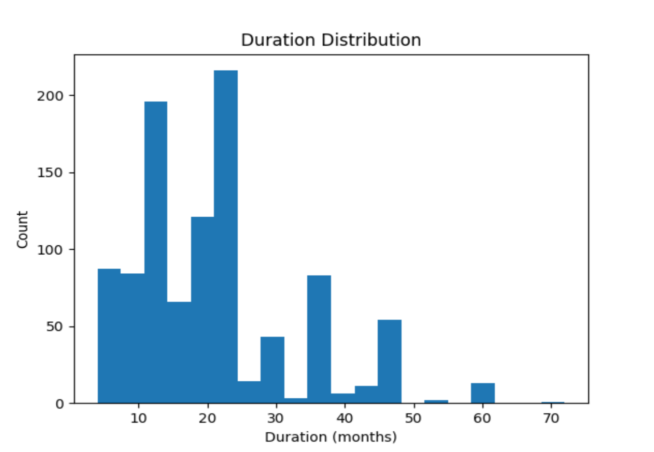
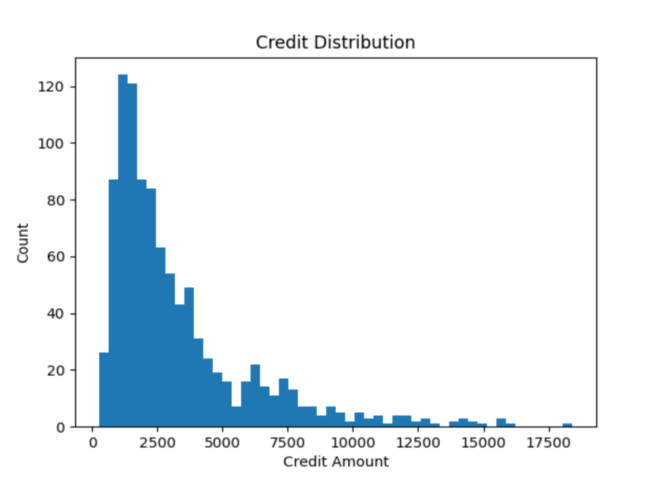
GERMAN CREDIT RISK ANALYSIS

A COMPREHENSIVE ANALYSIS OF FACTORS INFLUENCING CREDIT RISK

By Ethan Hartman







Data Overview

The dataset is of German loan borrowers.

- The information contained within the set relates to Individual characteristics (Sex, age, housing), account information (checking and savings account balance level), and loan information (purpose, duration, amount, and risk assessment).

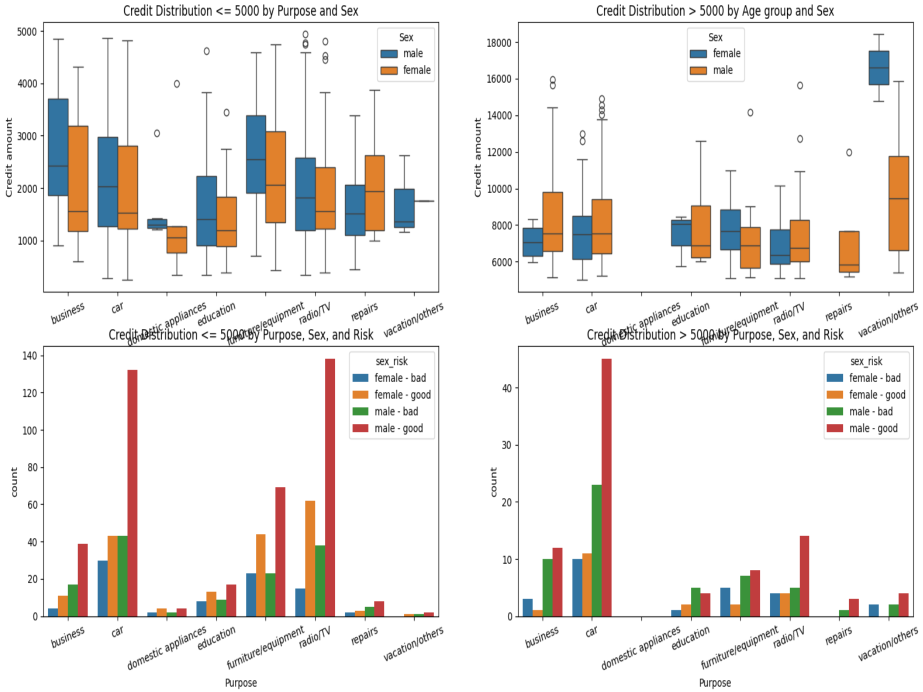
- Most of the data is categorical in nature but were converted to numerical identifiers later for correlation analysis.

- The Job column is numerical in nature, but information from the source of the dataset informs that it is categorical
  - ❖ 0 - unskilled non-resident, 1 - unskilled resident, 2 - skilled, 3 - highly skilled
  - ❖ This will be reflected in visualizations.

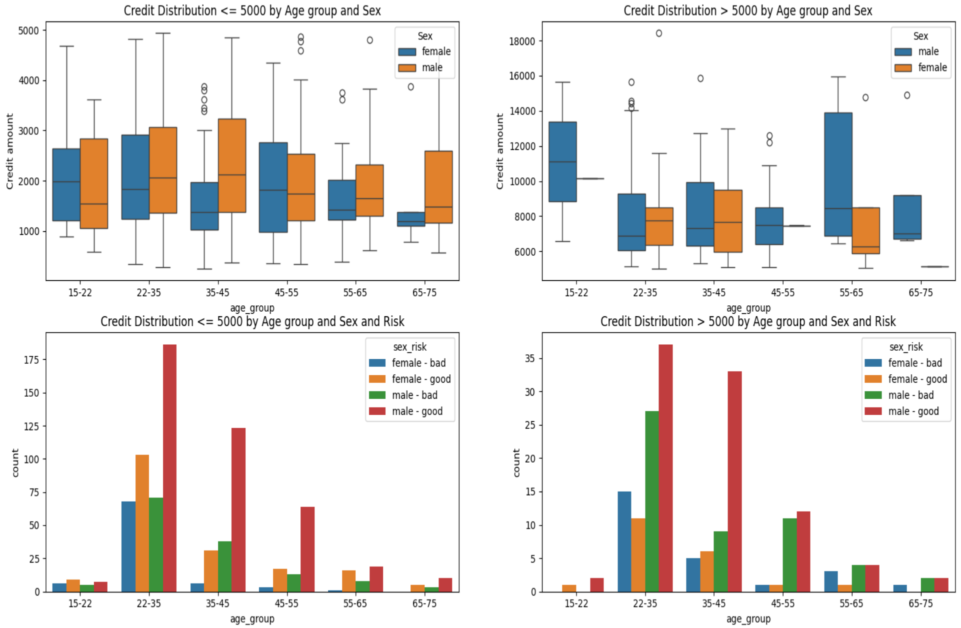
- Additionally, the only columns which contain null values are Savings and Checking. I These records were not dropped. Instead, I used an imputation method to fill them. This is discussed later.

CREDIT DEMOGRAPHICS

Credit distribution by Purpose, Sex, and Risk

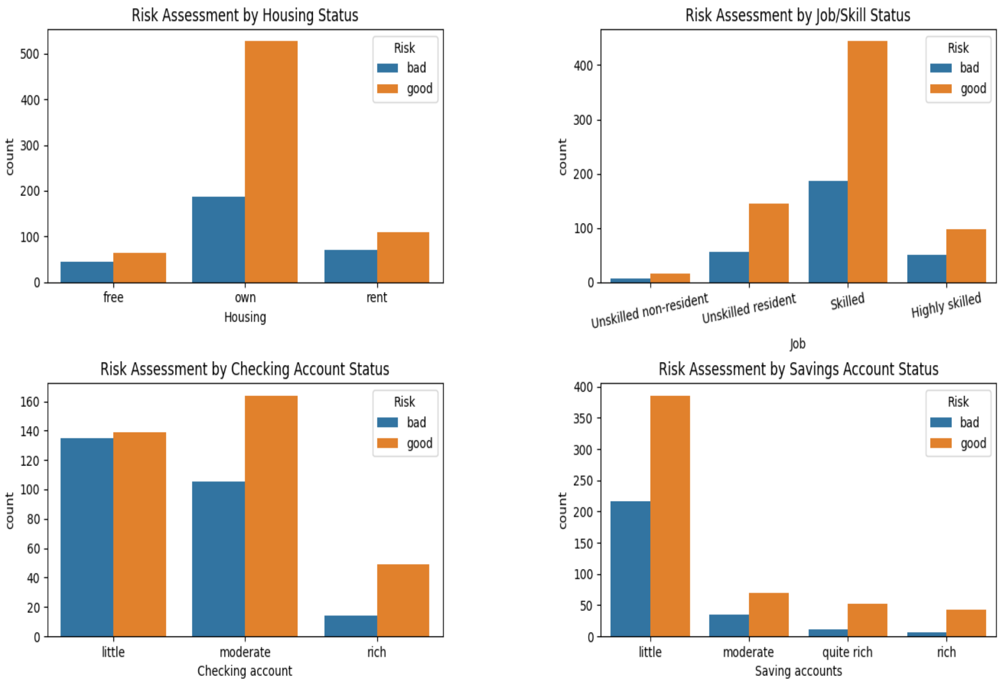


Distribution of Age, Credit Amount, and Duration



RISK ASSESSMENT BY DEMOGRAPHICS

Risk distribution by Housing, Job, Checking and Saving accounts



DATA PREPROCESSING AND FEATURE ENGINEERING

- NA'd values in the Checking and Saving accounts columns were filled by supplying the average value of the mean, median, and mode of their respective age group for that column.

- An upper and lower limit were calculated for numeric columns (Age, Credit, Duration) as follows:

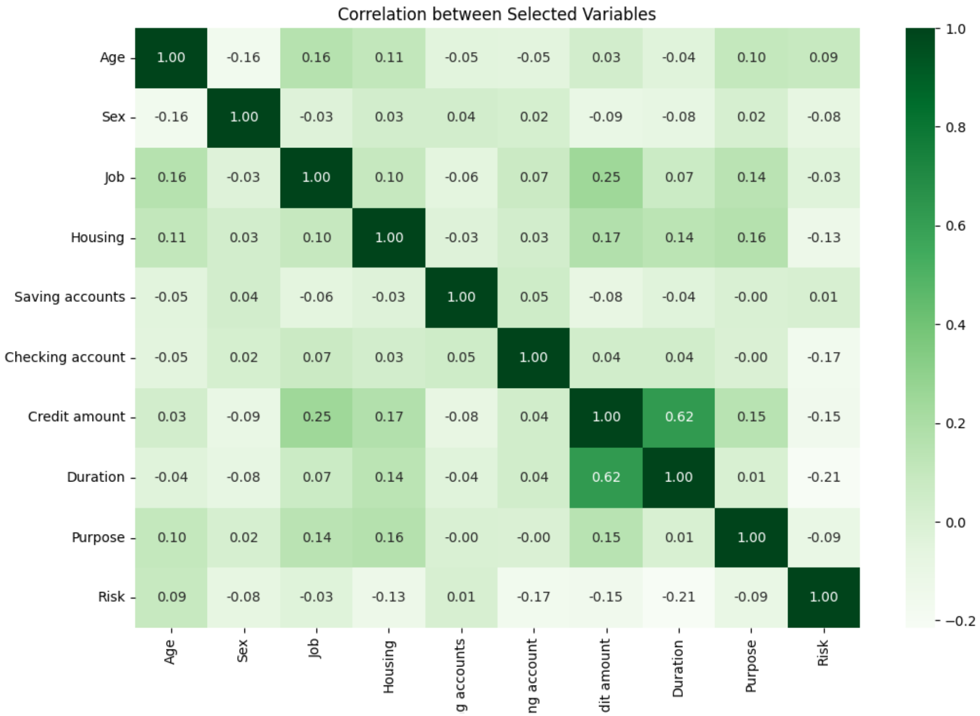
IQR = Q3(75% quartile) - Q1(25% quartile)

upper limit = Q3 + 1.5 * IQR , lower limit = Q1 - 1.5 * IQR

➢ Outliers were classified as values existing outside these limits. To maintain the dataset size and robustness, while reducing outlier influence, outlier values were changed to the appropriate limit value and the records were kept.

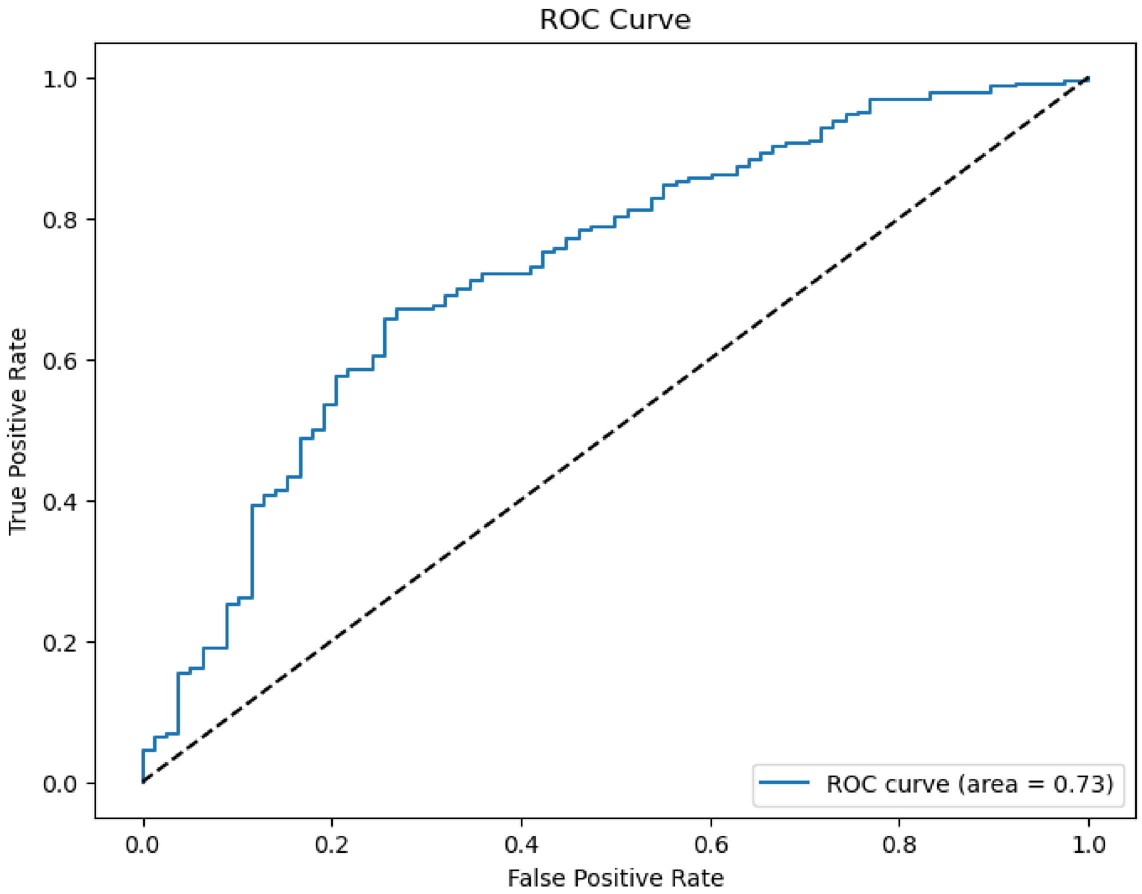
CORRELATION ANALYSIS

Correlation between Selected Variables



PREDICTIVE MODELING

Confusion Matrix and ROC Curve for Logistic Regression Model



	Predicted Bad Risk	Predicted Good Risk
Actual Bad Risk	18	60
Actual Good Risk	7	215

	Precision	Recall	F1-Score	Support
Bad Risk	0.72	0.23	0.35	78
Good Risk	0.78	0.97	0.87	222
Accuracy			0.78	300
Macro Avg	0.75	0.60	0.61	300
Weighted Avg	0.77	0.78	0.73	300